EXERCISE 1 A particle moves along a line with displacement function

$$s(t) = \frac{1}{4}t^2 - \ln(t+1), \quad t \geq 0, \quad s \text{ in metres, } t \text{ in seconds.}$$

1. **a.** Write an expression for the velocity of the particle at time t.
   b. Hence find when the particle is at rest.
2. **a.** Write an expression for the acceleration of the particle at time t.
   b. Hence show that velocity is never decreasing.

EXERCISE 2 A particle moves in a straight line with position relative to O given by $s(t) = 2t - \frac{4}{t+1}$cm, where $t \geq 0$ is the time in seconds.

1. Find velocity and acceleration functions for the particle's motion and draw sign diagrams for each of them.
2. Describe the motion of the particle at $t = 1$ second.
3. Does the particle ever change direction ? If so, where and when does it do this ?
4. Draw a diagram to illustrate the motion of the particle.
5. Find the time intervals when the :
   a. velocity is increasing
   b. speed is increasing

EXERCISE 3 A particle P moves in a straight line with position given by

$$s(t) = 80e^{-\frac{t}{10}} - 40t \quad \text{m},$$

where t is the time in seconds, $t \geq 0$.

1. Find the velocity and acceleration functions.
2. Find the initial position, velocity, and acceleration of P.
3. Discuss the velocity of P as $t \to \infty$.
4. Sketch the graph of the velocity function.
5. Find the exact time when the velocity is $-44\,\text{m s}^{-1}$.

EXERCISE 4 [SL students ONLY] A particle P moves in a straight line with position relative to the origin O given by

$$s(t) = 2t^3 - 9t^2 + 12t - 5 \quad \text{cm},$$

where t is the time in seconds, $t \geq 0$.

1. Find expressions for the particle's velocity and acceleration and draw sign diagrams for each of them.
2. Find the initial conditions.
3. Describe the motion of the particle at time $t = 2$ seconds.
4. Find the times and positions where the particle changes direction.
5. Draw a diagram to illustrate the motion of P.
6. Determine the time intervals when the particle's speed is increasing.

EXERCISE 5 [HL students ONLY] A point P moves in a straight line with velocity (m s^{-1}) given by

$$v(t) = e^{-t} - 8t^2e^{-2t}$$

at time t seconds, where $t \geq 0$.

1. Determine the first time t_1 at which P has zero velocity.

i. Find an expression for the acceleration of P at time t.

ii. Find the value of the acceleration of P at time t_1.